

Category A: Instruction Set Extensions (Easy)

A1. Multiply Instruction

MUL Xd, Xn, Xm

Reuse Lab 4 multiplier.

A2. Shift Instructions

LSL Xd, Xn, #imm

LSR Xd, Xn, #imm

Reuse Lab 3 shifter.

A3. Move Immediate

MOV Xd, #imm

Requires control logic modifications.

A4. Compare Instruction

CMP Xn, Xm

Updates ALU flags.

A5. Bitwise XOR

EOR Xd, Xn, Xm

Small ALU modification.

A6. NAND Instruction

NAND Xd, Xn, Xm

Simple ALU enhancement.

A7. Rotate Instructions

ROR Xd, Xn, #imm

ROL Xd, Xn, #imm

A8. Count Leading Zeros

CLZ Xd, Xn

A9. Arithmetic Shift Right

ASR Xd, Xn, #imm

A10. Increment / Decrement

INC Xd

DEC Xd

Category B: Branching and Control Flow (Medium)

B1. Conditional Branches

BEQ

BNE

BGT

BLT

Requires condition flags.

B2. Branch Prediction

Implement:

- Always Taken
- Always Not Taken

Compare prediction accuracy.

B3. Branch with Link

BL label

Stores return address.

B4. Subroutine Support

BL

RET

B5. Jump Register

BR Xn

B6. Loop Counter Hardware

Dedicated loop support instruction.

Category C: Memory System Extensions (Medium)

C1. Direct-Mapped Cache

Reuse Lab 6 concepts.

Requirements:

- Valid bit
- Tag comparison
- Cache hit detection

C2. Two-Way Set Associative Cache

C3. Memory-Mapped I/O

Reserve memory addresses for:

- LEDs
- Switches

Example:

```
STUR X1, [LED_ADDR]
```

C4. Stack Support

Implement:

PUSH

POP

C5. Stack Pointer Register

Dedicated SP register.

Category D: Floating Point Extensions (Medium-Hard)

D1. Floating Point Adder

Reuse Lab 4 floating-point design.

D2. Floating Point Multiplier

Reuse Lab 8 floating-point multiplier.

D3. Floating Point Register File

Separate FP register bank.

D4. Mixed Integer / Floating Point Execution

D5. Floating Point Compare

FCMP

Category E: Processor Architecture Extensions (Hard)

E1. Five-Stage Pipeline

Stages:

1. IF
2. ID
3. EX
4. MEM
5. WB

E2. Hazard Detection Unit

E3. Forwarding Unit

E4. Pipeline Stall Logic

E5. Multi-Cycle Multiplier

Category F: Research and Analysis Extensions

F1. Single-Cycle vs Pipeline Performance Study

F2. Cache Performance Analysis

Measure:

- Hit rate
- Miss rate

F3. Power Consumption Comparison

F4. Resource Utilization Analysis

Compare:

- ALU
- Multiplier
- Floating Point Unit

F5. Benchmark Program Suite

Create benchmark programs and analyze performance.

Extension G (Your choice)

You may choose an instructor approved extension of your choice